

Assignment:3

RAM and Storage Theory

Q1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

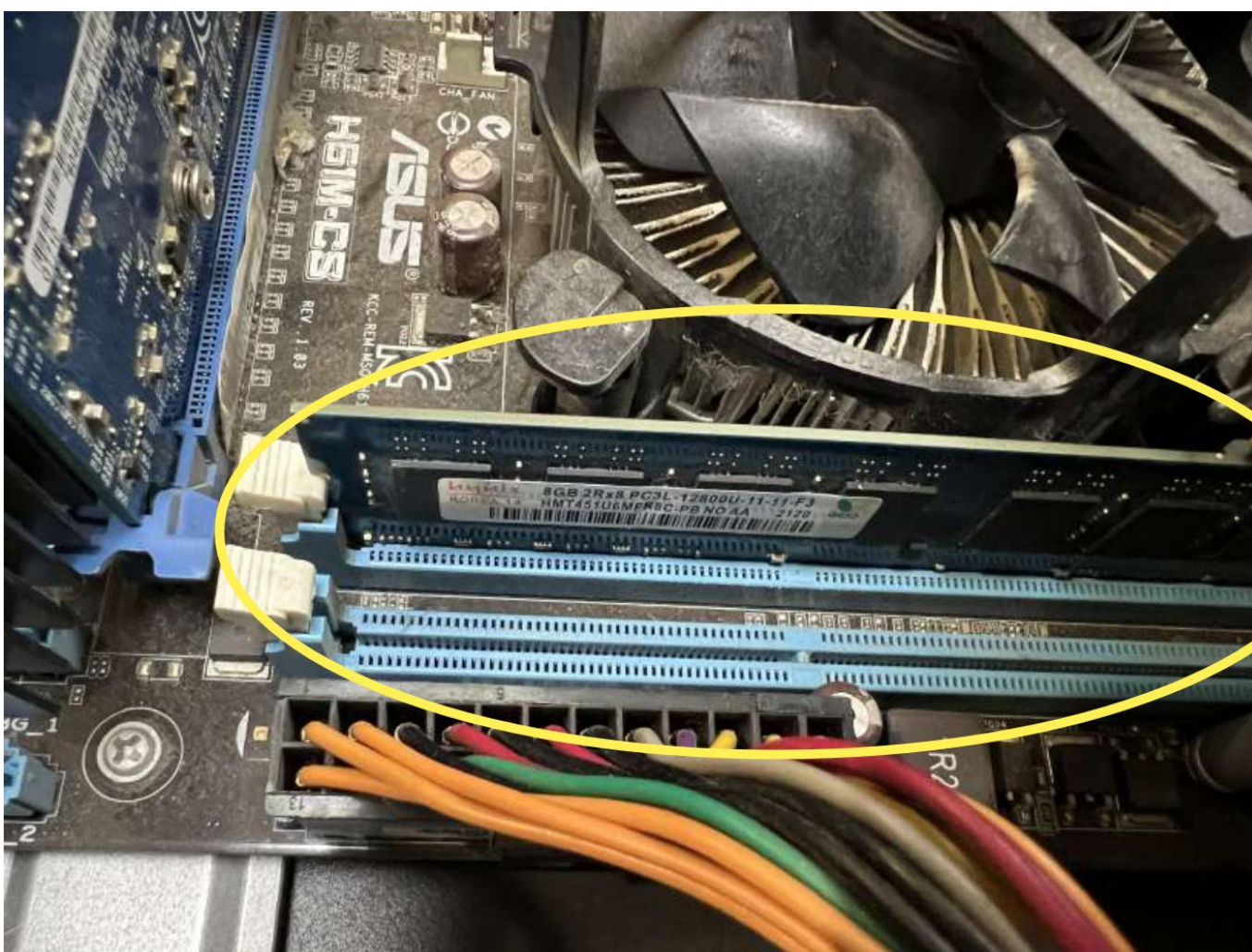
Ans.

Feature	DDR3	DDR4	DDR5
Release Year	2007	2014	2020
Speed	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
Voltage	1.5 V (1.35 V for DDR3L)	1.2 V	1.1 V
Performance	Good for basic computing	Faster and more efficient	Fastest with highest bandwidth
Power Consumption	Higher	Lower than DDR3	Lowest among the three
Capacity per Module	Up to 16 GB	Up to 64 GB	Up to 128 GB or more

**Q2.Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.**

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Q3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

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Feature	HDD	SSD	NVMe
Speed	Slow (80–160 MB/s)	Fast (500–550 MB/s)	Very Fast (2,000–7,000+ MB/s)
Technology	Uses spinning magnetic disks	Uses flash memory (SATA interface)	Uses flash memory with PCIe interface
Cost	Cheapest per GB	Moderate price	Most expensive per GB

Q4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

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❖ SATA Connector

